Technology Stack

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack

The following technology stack is used to develop the Rising Waters – Flood Prediction System Using Machine Learning.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Chart.js	Used to build a responsive and interactive web interface. Chart.js is used to visualize flood risk and prediction results through graphs.
2	Backend / Server-Side	Python, Flask	Flask provides a lightweight web framework for handling user requests, integrating the machine learning model, and generating predictions efficiently.
3	Database / Data Storage	CSV Dataset, Pandas DataFrame, Joblib (.pkl Model)	Historical flood data is stored in CSV format. Pandas is used for data processing, and Joblib stores the trained machine learning model for fast prediction.
4	Cloud / Hosting / Deployment	IBM Cloud / Local Flask Server	IBM Cloud provides scalable deployment for the application, while Flask enables easy local development and testing.
5	Version Control & CI/CD	Git, GitHub	Git is used for version control, and GitHub is used for collaboration, project management, and source code hosting.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib, VS Code	Scikit-learn is used for building the Machine Learning model, Pandas and NumPy for data preprocessing, Matplotlib for analysis, Joblib for model serialization, and VS Code as the development environment.

This technology stack is well suited for the Rising Waters project because it provides:

- A responsive web interface using HTML, CSS, and JavaScript
- Efficient backend processing with Python and Flask
- Accurate flood prediction using Scikit-learn Machine Learning
- Easy deployment on IBM Cloud
- Reliable version control using GitHub
- Fast and efficient data handling with Pandas, NumPy, and Joblib